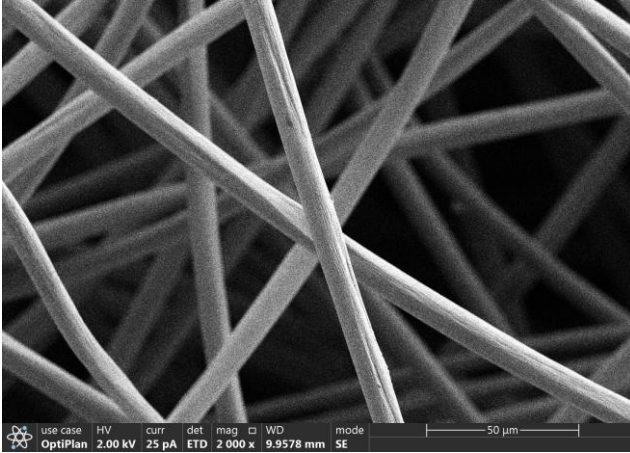


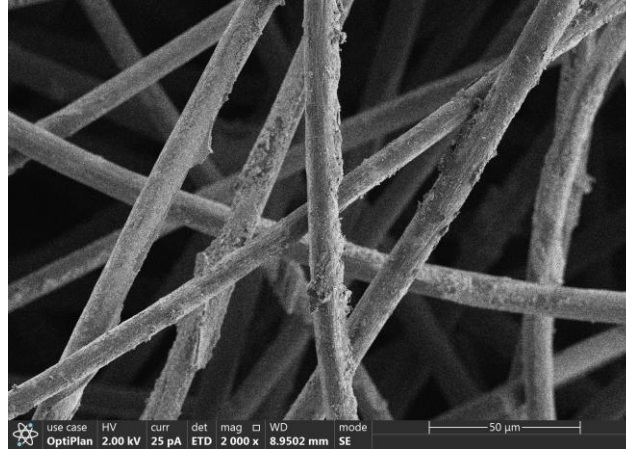
Introduction to Carbon Fiber Composite Materials Thermal Field Products

01

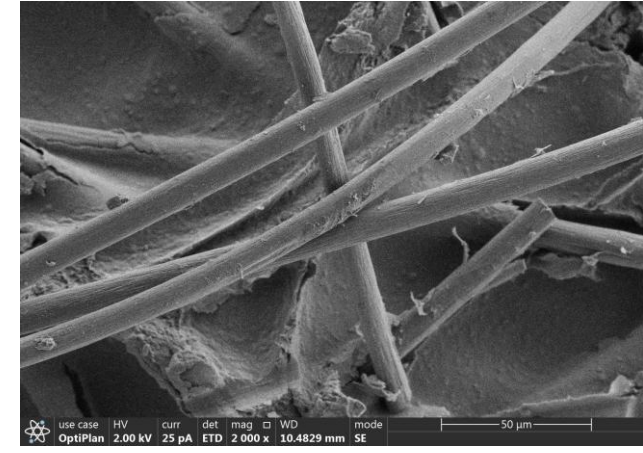
Material and Application



Soft felt



Cured felt



Carbon-carbon composite materials

Tailored thermal-field materials via continuous CVD and oxidation etching.

- Continuous CVD + oxidation etching: controlled fiber surface modification to final sealing
- Adjustable morphology, density and porosity
- At 1000 °C: thermal conductivity 0.15–35 W/m·K
- Compressive strength: 0.1–300 MPa
- At 600 °C in O₂: ablation rate 5–30%
- Process customization for specific customer environments

Maximum operating temperature of graphite product $\leq 3200^{\circ}\text{C}$

Metal heat treatment

$\leq 1500^{\circ}\text{C}$

Monocrystalline
silicon pulling

$\leq 1800^{\circ}\text{C}$

High temperature
graphitization treatment

$\leq 2200^{\circ}\text{C}$

SiC crystal
growth

$\leq 2400^{\circ}\text{C}$

02

Products

Soft felt

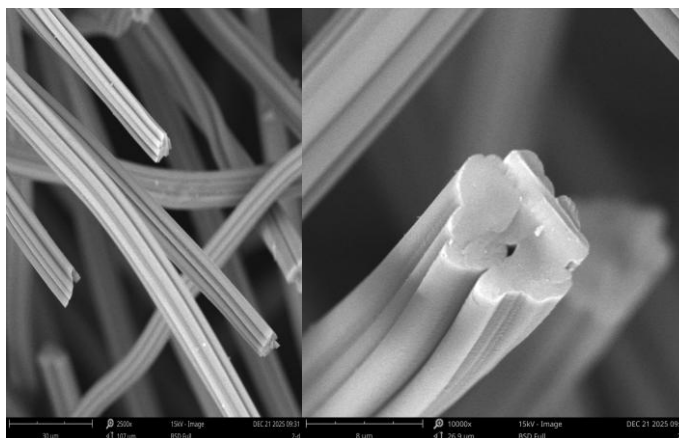


Name	Unit		Graphite soft felt		Remarks
			VLG-5/VLG-10	VLGY-5/VLGY-10	
Thickness	mm		5/10	5/10	Other specifications can be customized according to customer needs
Density	g/cm ³		0.08-0.11	0.07-0.09	
Width	m		1.3-1.6	1.3-1.6	
Breaking strength	Mpa		≥0.3	≥0.2	
Carbon content	%		≥99.80	≥99.80	
Thermal conductivity	W/m·k	1000℃	≤0.17	≤0.23	
		1500℃	≤0.35	≤0.38	
Heat treatment temperature	℃		≥1800	≥1800	
Fiber shedding rate	%		≤0.3	≤0.3	
Specific surface area	m ² /g		3-7	3-7	
Ash content	ppm		≤300	≤300	

Low-density viscose-based graphite felt for high-temperature insulation.

- Excellent high-temperature insulation material
- Low density and low thermal conductivity
- Ideal for PV crystal pulling and induction sintering furnaces
- Also suitable as ablation-resistant material
- Long-term service up to 3200 °C

Soft felt



Uniform needle-punched graphite felt,
configurable to customer specs.

- Needle punching delivers highly uniform insulation performance
- Available in different thickness, density, graphitization and fiber types
- Ash content: ≤ 77.06 ppm after purification

委托方: 嘉兴纳科新材料有限公司

检测方法: GDMS (辉光放电质谱法)

报告编号: P2208200502

测试日期: 2022-08-21

样品名称: 石墨软毡

样品编号: PF2300C

纯度: $C \geq 99.9923\%$

TMI: 77.06 ppmat

GDMS 数据

元素	含量	元素	含量
Li	< 0.05	Ag	< 0.05
Be	< 0.05	Cd	< 0.1
B	1.8	In	辅助电极
C	主成分	Sn	< 1
F	1.3	Sb	< 0.05
Na	0.16	Te	< 0.1
Mg	1.5	I	< 100
Al	10	Cs	< 0.05
Si	45	Ba	1.2
P	0.29	La	< 5.0
S	5.9	Ce	< 0.5
Cl	1.4	Pr	< 0.5
K	2.5	Nd	< 0.05
Ca	5.3	Sm	< 0.05
Sc	< 0.05	Eu	< 0.05
Ti	1.3	Gd	< 0.05
V	0.13	Tb	< 0.05
Cr	< 0.1	Dy	< 0.05
Mn	< 0.05	Ho	< 0.05
Fe	0.33	Er	< 0.05
Co	< 0.05	Tm	< 0.05
Ni	< 0.05	Yb	< 0.05
Cu	< 0.05	Lu	< 0.05
Zn	< 0.05	Hf	< 0.05
Ga	< 0.05	Ta	< 50
Ge	< 0.1	W	< 0.05
As	< 0.1	Re	< 0.05
Se	< 0.1	Os	< 0.05
Br	< 0.1	Ir	< 0.05
Rb	< 0.1	Pt	< 0.1
Sr	< 0.1	Au	< 0.1
Y	< 0.05	Hg	< 0.5
Zr	0.09	Tl	< 0.5
Nb	< 0.1	Pb	< 0.5
Mo	0.34	Bi	< 0.05
Sn	< 0.05	Th	< 0.01
PH	< 0.05	U	0.02
PD	< 0.05		

责任声明:

本检测报告数据基于实验室检测数据, 不代表任何第三方检测结果, 样品编号为客户提供。

编制人: 孙永发

审核人: 孙永发

+86 512 81880919

info@patlabs.com.cn
www.patlabs.com.cn

江苏省苏州工业园区东长路88号F1幢1F101室
F1-1F101, 88 Dongchang Rd, Suzhou Industrial Park, Jiangsu

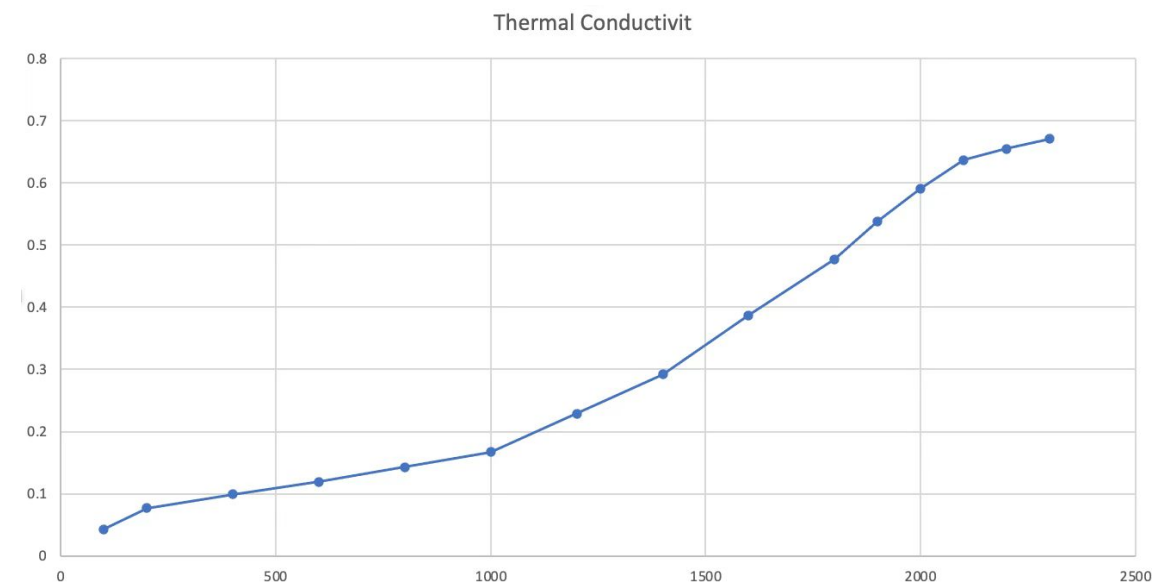
Thermal insulation comparison

Product	Thermal conductivity (1000°C) (W/m·k)	Density (g/cm ³)	Air Ablativity (600°C, 1h) (%)
VLG-10 China VLG-10 soft felt	0.17	0.1	6.21
VLGY-10 China VLGY-10 soft felt	0.23	0.08	9.95
SGL German SGL soft felt	0.25	0.1	10.12
French Melson soft felt	0.25	0.08	6.32
American Morgan soft felt	0.23	0.1	10.22

Client-validated insulation performance with measurable ROI.

- Test methods: DRX-I-SPB (thermal conductivity) | Muffle furnace (air erosion)
- Soft felt comparison: better insulation → lower heat loss
- ~20% energy saving | ~50% material cost saving | ~7% total cost reduction

Actual results depend on local conditions.



DRX-I-SPB Thermal Conductivity Tester for Measuring Thermal Conductivity



Determination of air erosion rate using muffle furnace

C/C composite materials

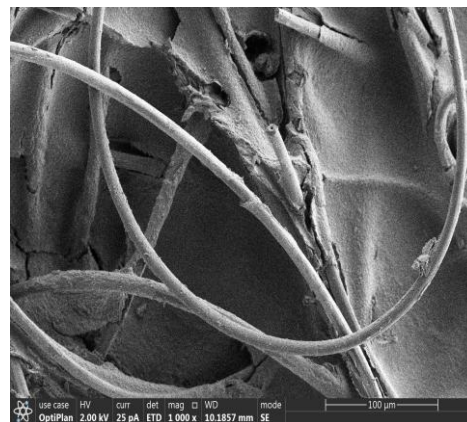


Name	Unit	C/C composite material	Remarks
Thickness	mm	1.2-200	Other specifications can be customized according to customer needs
Density	g/cm ³	≤1.8	
Width/Diameter	mm	≤1800	
Length/Height	mm	≤2400	
Extension strength	Mpa	≥90	
Compressive strength	Mpa	≥120	
Thermal conductivity	W/m·k	1000°C	
		1500°C	
Carbon content	%	≥99.80	
Heat treatment temperature	°C	≥1800	
Ash content	ppm	≤200	

Lightweight, high-temperature C/C composites for thermal structures.

- Carbon fiber reinforcement + pyrolytic/resin/asphalt carbon matrix
- Low density (1.65–2.0 g/cm³), low CTE, high specific strength
- Mechanical properties increase with temperature up to 2200 °C
- Excellent ablation resistance, thermal shock resistance and fatigue resistance
- Adjustable friction coefficient (0.2–0.45); long service life
- Applications: PV/semiconductor, metallurgy, heat treatment, friction and structural parts
- Shapes: plates, discs, cylinders, special forms, bolts and fasteners

C/C composite materials



Flexible C/C production with precise density and surface treatment.

- Unique gas-liquid carburizing densification for convenient density control
- Custom textile and needle-punching processes for different operating conditions
- Optional boronizing or siliconizing on request
- Ash content: 4.98 ppm after purification

委托方: 嘉兴纳科新材料有限公司

检测方法: GDMS (辉光放电质谱法)

GDMS 数据

报告编号: P2208230801

测试日期: 2022-08-24

样品名称: 碳碳样块

样品编号: 1#

纯度: C=99.99950%

灰分: 4.98 ppmwt

元素	含量	元素	含量
Li	< 0.01	Ag	< 0.05
Be	< 0.05	Cd	< 0.05
B	0.65	In	< 0.05
C	主成分	Sn	< 0.05
F	< 0.5	Sb	< 0.05
Na	< 0.05	Te	< 0.05
Mg	< 0.05	I	< 0.05
Al	< 0.05	Cs	< 0.01
Si	2	Ba	< 0.05
P	< 0.1	La	< 0.05
S	0.16	Ce	< 0.05
Cl	0.07	Pr	< 0.01
K	< 0.5	Nd	< 0.01
Ca	< 0.05	Sm	< 0.01
Sc	< 0.01	Eu	< 0.01
Ti	1.4	Gd	< 0.01
V	0.66	Tb	< 0.01
Cr	< 0.1	Dy	< 0.01
Mn	< 0.01	Ho	< 0.01
Fe	0.04	Er	< 0.01
Co	< 0.01	Tm	< 0.01
Ni	< 0.05	Yb	< 0.01
Cu	< 0.05	Lu	< 0.01
Zn	< 0.05	Hf	< 0.01
Ga	< 0.01	Ta	< 100
Ge	< 0.05	W	< 0.05
As	< 0.05	Re	< 0.01
Se	< 0.5	Os	< 0.01
Br	< 0.05	Ir	< 0.01
Pb	< 0.05	Pt	< 0.05
Sr	< 0.05	Au	< 0.5
Y	< 0.05	Hg	< 0.1
Zr	< 0.05	Tl	< 0.05
Nb	< 0.05	Pb	< 0.05
Mo	< 0.05	Bi	< 0.05
Ru	< 0.05	Th	< 0.01
Rh	< 0.01	U	< 0.01
Pd	< 0.05		

免责声明:
未经我方书面批准,不得部分复制本报告。
本报告仅对C/C样品检测结果负责。
样品编号为客户提供。



编制人: 孙永发

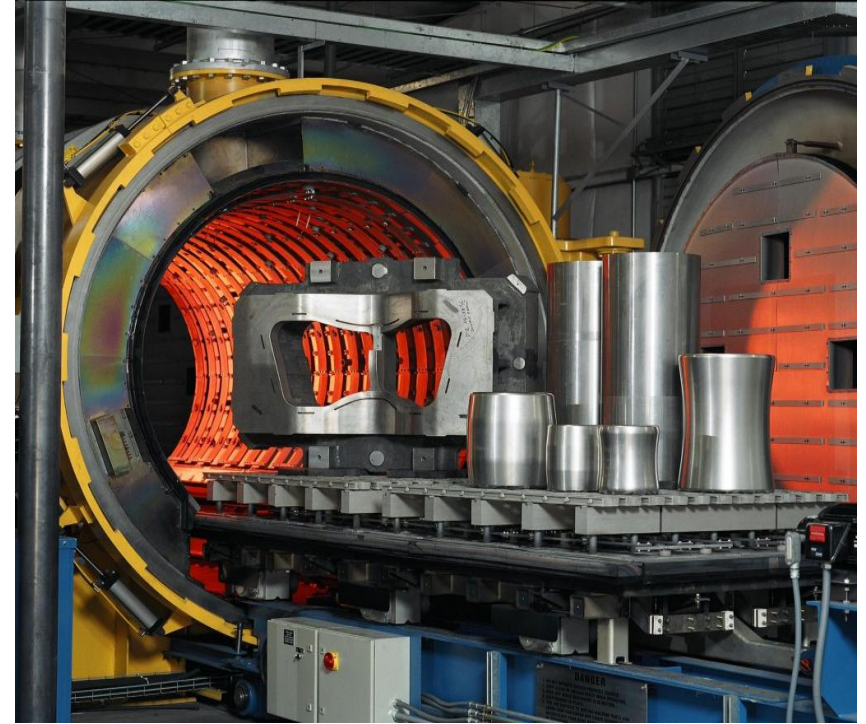
审核人: 孙永发

+86 512 81880919

info@patlabs.com.cn
www.patlabs.com.cn

江苏省苏州工业园区东长路88号F1幢1F101室
F1-1F101, 88 Dongchang Rd, Suzhou Industrial Park, Jiangsu

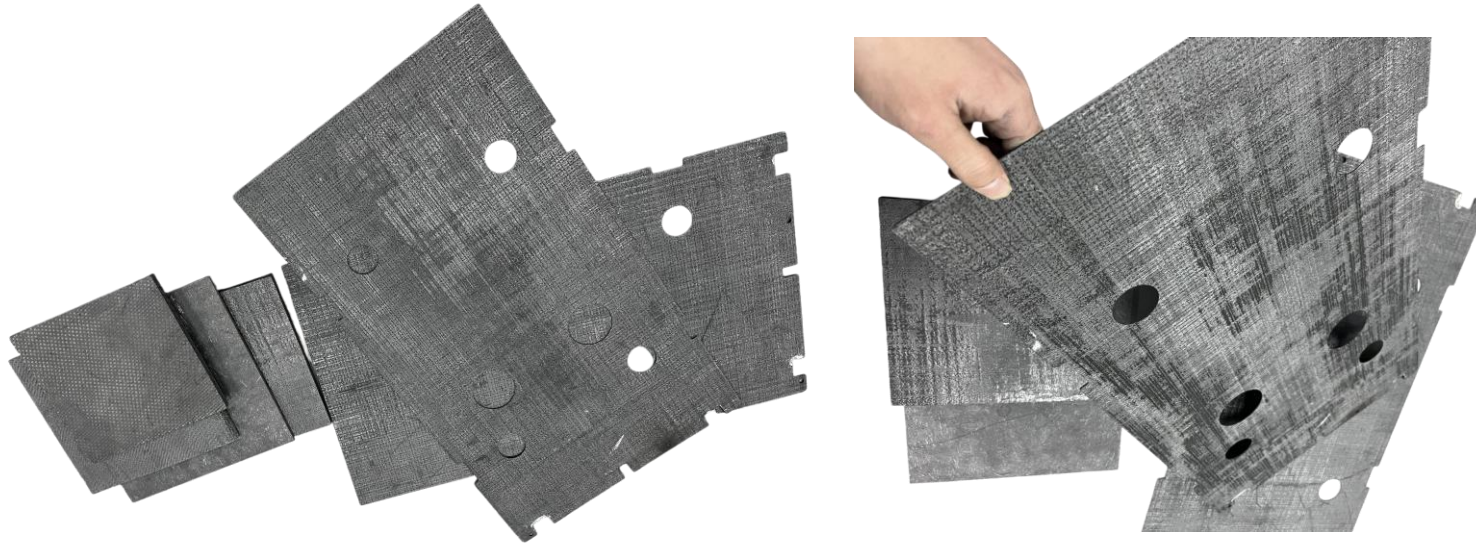
C/C composite — heat treatment racks and shelves



Lightweight C/C racks with near-zero thermal expansion.

- High-density C/C racks assembled by machining and splicing
- CTE $\sim 0.5 \times 10^{-6}/^{\circ}\text{C}$ vs. stainless steel $10\text{--}17 \times 10^{-6}/^{\circ}\text{C}$
- Service up to 3200°C in oxygen-free environment
- Density $\sim 1.5 \text{ g/cm}^3$: lightweight, easy stacking and assembly
- Carbon content $\geq 99.9\%$: chemically inert, no reaction with metal workpieces

C/C composite — heat-treated carrier plate



Crack-resistant C/C carrier plates for demanding heat treatment.

- Lower density (1.5 g/cm^3) than mullite/SiC; higher strength than graphite
- Compressive strength $\geq 180 \text{ MPa}$; flexural strength $\geq 120 \text{ MPa}$
- High toughness: no cracking under impact
- Better oxidation resistance and longer life than graphite
- Graphitization-related lubricity protects workpiece surfaces
- Suitable for diverse heat treatment environments up to $3200 \text{ }^\circ\text{C}$ (inert)



Cured felt

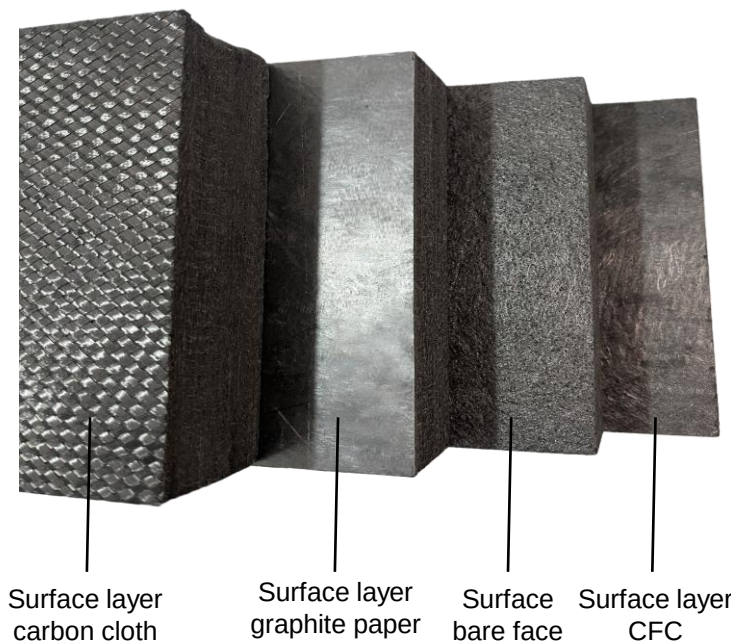


Name	Unit		Cured felt	Remarks
Thickness	mm		20-300	Other specifications can be customized according to customer needs
Density	g/cm ³		≤0.18	
Width/Diameter	mm		≤1800	
Length/Height	mm		≤2400	
Extension strength	Mpa		≥1	
Compressive strength	Mpa		≥2	
Thermal conductivity	W/m·k	1000°C	≤0.25	
		1500°C	≤0.30	
Carbon content	%		≥99.80	
Heat treatment temperature	°C		≥1800	
Ash content	ppm		≤300	

Rigid, self-supporting hard insulation for high-temperature furnaces.

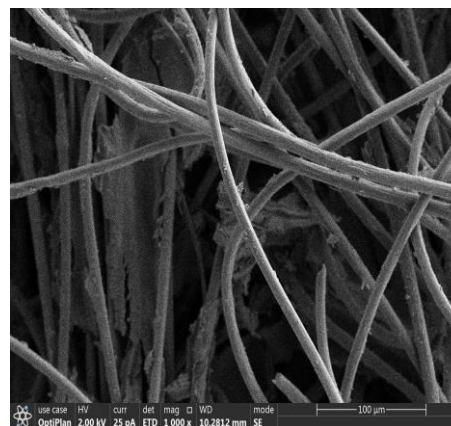
- Carbon fiber prefabricate, resin infiltration, curing and graphitization (>1800 °C)
- Rigid and self-supporting; low specific gravity; no short-fiber shedding
- Low thermal conductivity; service temperature ≤3200 °C
- For gas quenching, pressure sintering, SiC recrystallization, mono/poly-Si furnaces
- Shapes: flat plate, disc, cylinder; long service life; easy installation

Cured felt



Configurable cured felt variants and surface coatings.

- Types: integral needle-punched, laminated soft felt, short-fiber wet-formed
- Surface options: carbon cloth, graphite paper, bare face, CFC board
- Custom density balancing oxidation resistance and thermal insulation
- Ash content: 39.5 ppm after purification



委托方: 嘉兴纳科新材料有限公司

检测方法: GDMS (辉光放电质谱法)

GDMS 数据

单位: ppm wt

报告编号: P2208011601

测试日期: 2022-08-03

样品名称: 硬毡

样品编号: 纯化硬毡

纯度: C=99.9961%

TMI: 39.5 ppmwt

元素	含量	元素	含量
Li	< 0.05	Ag	< 0.05
Be	< 0.05	Cd	< 0.5
B	0.02	In	辅助电极
C	主成分	Sn	< 1
F	2	Sb	< 0.05
Na	< 0.05	Te	< 0.05
Mg	0.45	I	< 100
Al	2.2	Cs	< 0.05
Si	15	Ba	< 0.1
P	1.1	La	< 50
S	4.6	Ce	< 0.5
Cl	2.6	Pr	< 0.1
K	1.1	Nd	< 0.05
Ca	8.6	Sm	< 0.05
Sc	< 0.05	Eu	< 0.5
Ti	0.75	Gd	< 0.05
V	< 0.01	Tb	< 0.05
Cr	< 0.1	Dy	< 0.05
Mn	< 0.05	Ho	< 0.05
Fe	0.9	Er	< 0.05
Co	< 0.05	Tm	< 0.05
Ni	< 0.05	Yb	< 0.05
Cu	< 0.05	Lu	< 0.05
Zn	< 0.1	Hf	< 0.05
Ga	< 0.05	Ta	< 10
Ge	< 0.1	W	0.18
As	< 0.1	Re	< 0.05
Br	< 0.5	Os	< 0.05
Rb	< 0.1	Ir	< 0.05
Sr	< 0.1	Pt	< 0.1
Y	< 0.05	Au	< 0.1
Zr	< 0.05	Hg	< 0.5
Nb	< 0.1	Tl	< 0.5
Mo	< 0.05	Pb	< 0.5
Ru	< 0.05	Bi	< 0.05
Rh	< 0.05	Th	< 0.01
Pd	< 0.05	U	< 0.01

责任声明:
未经我方书面批准不得部分复制本报告。
本检测报告仅对取得的测试样品负责。
样品编号为客户提供。



编制人: 孙承贤

审核人: 孙承贤

31880919

info@patlabs.com.cn
www.patlabs.com.cn

江苏省苏州工业园区东长路88号F1幢1F101室
F1-1F101, 88 Dongchang Rd, Suzhou Industrial Park, Jiangsu

NEEA-2023-0780

第: 001/013

Thank you for your attention